

Tutorial-9

Q 1: Compute the Jordan canonical form of following matrices.

$$(A) \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -2 \\ 0 & 1 & 3 \end{pmatrix}$$

Solⁿ Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -2 \\ 0 & 1 & 3 \end{pmatrix}$

$$\text{Char}(A) = \det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 0 & 0 \\ 0 & -\lambda & -2 \\ 0 & 1 & 3-\lambda \end{pmatrix}$$

$$= (1-\lambda)[(-\lambda)(3-\lambda)+2] = 0$$

$$= \lambda = 1, 2, 1$$

$$\text{Char}(A) = (\lambda-1)^2(\lambda-2)$$

$$\text{min}(A) = (\lambda-1)(\lambda-2)$$

$$\text{partition} = 2+1+1$$

So J.C.F = $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$$(B) \begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Soln
E.V = 1, 1, 1, 1

$$\text{Char}(A) = (x-1)^4$$

$$\text{min}(A) = (x-1)^4$$

$$J.C.F, \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$(C) \begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & -1 \\ -6 & -5 & 3 \end{pmatrix}$$

Soln
Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 2 & 1 & -1 \\ -6 & -5 & 3 \end{pmatrix}$

$$\text{Char } A = \lambda^3 - 4\lambda^2 + [-2+6-2]\lambda - [-4] = 0$$

$$\lambda^3 - 4\lambda^2 + 2\lambda + 4 = 0$$

$$\lambda = 2, 1+\sqrt{3}, 1-\sqrt{3}$$

$$\text{Char}(A) = (x-2)(x-1+\sqrt{3})(x-1-\sqrt{3}) = \text{min } A$$

$$\text{so } J.C.F = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1+\sqrt{3} & 0 \\ 0 & 0 & 1-\sqrt{3} \end{bmatrix}$$

Q2:- Let $T \in M_{n \times n}(\mathbb{C})$ be nilpotent matrix with nilpotent ~~matrix~~ index n . Find a matrix P s.t.

$$P^{-1}TP = \begin{bmatrix} 0 & 1 & & 0 \\ & 0 & \ddots & \\ & & 0 & 1 \\ 0 & & & 0 \end{bmatrix}_{n \times n}$$

Soln

$$n(T) = n$$

$$T^{n-1} \neq 0$$

$$\exists x \neq 0 \text{ s.t. } T^{n-1}x \neq 0$$

$$\{x, Tx, T^2x, \dots, T^{n-1}x\} \text{ is L.B.}$$

$$P = [T^{n-1}x, T^{n-2}x, T^{n-3}x, \dots, Tx, 0x]$$

Q// Use the ques in tutorial 8; if A is 2×2 complex matrix with complex entries then A is similar to a matrix one of the two types.

$$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \quad \begin{bmatrix} a & 0 \\ 1 & a \end{bmatrix}$$

Soln If A is diagonalizable (distinct eigenvalues) then $A \sim \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ where a, b are the eigen values of A .

If A has same eigenvalue say a then $\text{Ch}_A(x) = (x-a)^2$

$$\text{So, } (A-a)^2 = 0$$

$$\text{by exercise, } A-a \sim \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$A \sim \begin{bmatrix} a & 0 \\ 1 & a \end{bmatrix}$$

Q3:- Let α and β be distinct complex numbers. Let $J_r(x)$ be the Jordan block associated to λ with size r . If a matrix T is similar to

$$\begin{bmatrix} J_{n_1}(\alpha) & & & \\ & J_{n_2}(\alpha) & & \\ & & J_{m_1}(\beta) & \\ & & & J_{m_2}(\beta) \end{bmatrix}$$

Find the minimal polynomial for T .

Soln:- $m_T(x) = (x-\alpha)^{\max(n_1, n_2)} (x-\beta)^{\max(m_1, m_2)}$

Q4:- Let $A \in M_{5 \times 5}(\mathbb{C})$. If the minimal polynomial of A is $(x-1)^3(x-2)^2$. then find the Jordan form of A .

Soln:- $m_A(x) = (x-1)^3(x-2)^2$

Jos. $\left[\begin{array}{ccc|cc} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ \hline 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 2 \end{array} \right] \quad 3+2$

Q5: Let A be a matrix with
Char. pol. $(x-1)^3(x-2)^2$. Find all
possible Jordan forms of A .

Soln: $\text{Char}(A) = (x-1)^3(x-2)^2$

- Choices of $\text{min}(A) =$
- (i) $(x-1)^3(x-2)^2$ ✓
 - (ii) $(x-1)^2(x-2)^2$
 - (iii) $(x-1)^3(x-2)$
 - (iv) $(x-1)^2(x-2)^2$
 - (v) $(x-1)^3(x-2)$
 - (vi) $(x-1)^2(x-2)$

total choices for J.C.F = $p(n_1) \cdot p(n_2)$
 $\Rightarrow p(3) \cdot p(2)$
 $\Rightarrow 2 \cdot 2$
 $\Rightarrow 4$

(i) J.C.F $\begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$

(iii) $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$

(iv) $\begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$

(v) $\begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$

(vi) $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$

Tutorial X

Q1: Let V be an IPS. For any $x, y \in V$

Show that

$$\langle x, y \rangle = \frac{1}{4} \sum_{k=0}^3 i^k \|x + i^k y\|^2$$

$$\underline{\text{Soln:}} \quad \frac{1}{4} \sum_{k=0}^3 i^k \|x + i^k y\|^2$$

$$= \frac{1}{4} \sum_{k=0}^3 i^k \langle x + i^k y, x + i^k y \rangle$$

$$= \frac{1}{4} \sum_{k=0}^3 [i^k \langle x, x + i^k y \rangle + i^k \cdot i^k \langle y, x + i^k y \rangle]$$

$$= \frac{1}{4} \sum_{k=0}^3 [\langle x, y \rangle + i^k \langle y, y \rangle + i^k \langle x, y \rangle + i^{2k} \langle y, y \rangle]$$

$$= \frac{1}{4} [\langle x, y \rangle + \langle y, y \rangle + \langle x, y \rangle + \langle y, y \rangle +$$

$$\langle x, y \rangle + i \langle y, y \rangle + i \langle x, y \rangle - \langle y, y \rangle$$

$$+ \langle x, y \rangle - \langle y, y \rangle - \langle x, y \rangle + \langle y, y \rangle + \langle x, y \rangle$$

$$- i \langle y, y \rangle - i \langle x, y \rangle - \langle y, y \rangle]$$

$$= \langle x, y \rangle.$$

Q2: Let V be an IPS with an ONB

$\{x_1, x_2, \dots, x_n\}$. If $\alpha_1, \alpha_2, \dots, \alpha_n$ are scalars

find $x \in V$ s.t. $\langle x, x_i \rangle = \alpha_i \quad \forall i=1, 2, \dots, n$.

Is such x unique?

Soln: Let $x \in V$

s.t. $\langle x, x_i \rangle = \alpha_i \quad \forall i=1, 2, \dots, n$

As $x \in V$, $x = \sum_{j=1}^n \beta_j x_j$

$$\text{So, } \left\langle \sum_{j=1}^n \beta_j x_j, x_i \right\rangle = \alpha_i$$

$$\Rightarrow \sum_{j=1}^n \beta_j \langle x_j, x_i \rangle = \alpha_i$$

Since $\{x_1, x_2, \dots, x_n\}$ is ONB

so $\langle x_i, x_j \rangle = 0$ if $i \neq j$

$$\Rightarrow \beta_i \langle x_i, x_i \rangle = \alpha_i$$

$$\Rightarrow \beta_i = \alpha_i \quad \forall i \quad (\because \langle x_i, x_i \rangle = 1)$$

$$\text{Thus, } x = \sum_{j=1}^n \alpha_j x_j$$

x is fixed.

Q3: (A) Show that $\langle A, B \rangle = \text{tr}(AB^*)$ defines an inner product space on $M_{n \times n}(\mathbb{C})$.

Soln: (i) linearity:

$$\begin{aligned}\langle A+B, C \rangle &= \text{tr}((A+B)C^*) \\ &= \text{tr}(AC^* + BC^*) \\ &= \text{tr}(AC^*) + \text{tr}(BC^*) \\ &= \langle A, C \rangle + \langle B, C \rangle\end{aligned}$$

$$\begin{aligned}\text{(ii)} \quad \langle \alpha A, B \rangle &= \text{tr}(\alpha AB^*) \\ &= \alpha \text{tr}(AB^*) \\ &= \alpha \langle A, B \rangle\end{aligned}$$

$$\begin{aligned}\text{(iii)} \quad \langle A, B \rangle &= \text{tr}(AB^*) \\ \text{Let } B^* &= (b_{ij}), A = (a_{ij}) \\ B^* &= C = (c_{ij}) \\ D &= AB^* = (d_{ij})\end{aligned}$$

$$\text{then } d_{ik} = \sum_j a_{ij} \bar{c}_{jk}$$

$$\text{tr}(AB^*) = \sum_i d_{ii} = \sum_i \sum_j a_{ij} \bar{c}_{ji} = \sum_i \sum_j a_{ij} \bar{b}_{ji}$$

N.w.,

$$\begin{aligned}\langle B, A \rangle &= \text{tr}(BA^*) \\ \text{Let } A &= (a_{ij}), B = (b_{ij}) \\ A^* &= E = (\bar{a}_{ji}) \\ BA^* &= F = (f_{ij})\end{aligned}$$

$$f_{ik} = \sum_j b_{ij} \bar{a}_{jk}$$

$$\begin{aligned}\text{tr}(BA^*) &= \text{tr}(F) = \sum_i f_{ii} \\ &= \sum_i \sum_j b_{ij} \bar{a}_{ji}\end{aligned}$$

$$\begin{aligned}\therefore \langle B, A \rangle &= \left(\sum_i \sum_j b_{ij} \bar{a}_{ji} \right) \\ &= \sum_i \sum_j \bar{a}_{ji} b_{ij} = \langle A, B \rangle\end{aligned}$$

$$\begin{aligned}\text{(iv)} \quad \langle A, A \rangle, A \neq 0 \\ \Rightarrow \text{tr}(AA^*) \\ \Rightarrow \sum_{i,j} a_{ij} \bar{a}_{ji} \\ \neq 0.\end{aligned}$$

(B) With the above inner product on $M_{n \times n}(\mathbb{C})$, find the orthocomplement of the subspace of diagonal matrix.

Soln:

Let $D =$ Subspace of all diagonal matrix

$$D = \left\{ A \in M_{n \times n}(\mathbb{C}) \mid A_{ij} = 0 \text{ for } i \neq j \right\}$$

Let $D^\perp$ is orthocomplement of subspace D

Let $B \in D^\perp$

then $\text{tr}(AB^*) = 0$ for all $A \in D$

$$\text{tr}(AB^*) = 0$$

$$\sum_{i=1}^n a_{ii} \overline{b_{ii}} = 0$$

$$\Rightarrow b_{ii} = 0 \quad \forall i = 1, 2, \dots, n.$$

hence $D = \{ B \in M_{n \times n}(\mathbb{C}) \mid B_{ij} = 0 \quad \forall i = j \}$

$$V = \{ M_{n \times n}(\mathbb{C}) \}$$

Let $\langle A, B \rangle = \text{tr}(AB^*)$ be an IPS. Let P be an invertible matrix in V , and T_P is a linear operator on V defined by

$$T_P(A) = P^* A P. \text{ Find adjoint of } T_P.$$

Soln:

for any $A \in M_{n \times n}(\mathbb{C})$

$$\langle T_P^*(A), B \rangle = \langle A, T_P(B) \rangle$$

$$= \langle A, P^* B P \rangle$$

$$= \text{tr}(A(P^* B P)^*)$$

$$= \text{tr}(A P^* B^* (P^*)^*)$$

$$= \text{tr}(A P^* B^* (P^*)^{-1})$$

$$= \text{tr}((P^*)^{-1} A P^* B^*)$$

$$\text{We obtain } \langle T_P^*(A), B \rangle = \langle (P^*)^{-1} A P^*, B \rangle$$

$$T_P^*(A) = (P^*)^{-1} A P^*$$

$\square$

6 determine whether the following matrices are normal

(i) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, (ii) $B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

Find unitary matrix U_1 and U_2 s.t.

$U_1^* A U_1$ and $U_2^* B U_2$ are diagonal.

Step 1: (i) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$$A^* = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A A^* = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$A^* A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

Therefore $AA^* = A^*A$, A is normal.

Step 2: find eigenvalue and eigenvector of A .

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = 0$$

$$= (1-\lambda)^2 - 1 = 0$$

$$\lambda = 0, 2$$

for $\lambda = 0$,

$$AX = \lambda X$$

$$AX = 0$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$x_1 + x_2 = 0$$

$$x_1 = -x_2$$

$$U_1 = \begin{pmatrix} -x_2 \\ x_2 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

for $\lambda = 2$

$$AX = \lambda X$$

$$AX = 2X$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$x_1 + x_2 = 2x_1$$

$$x_1 + x_2 = 2x_2$$

$$x_1 = x_2$$

$$U_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Step 3: find unitary matrix U_1

$$\|U_1\|^2 = \sqrt{2}$$

$$\|U_2\|^2 = \sqrt{2}$$

$$U_1 = \left[\frac{U_1}{\|U_1\|^2}, \frac{U_2}{\|U_2\|^2} \right] = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

step 4 verify $U^*AU = I$

$$= \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 & \sqrt{2} \\ 0 & \sqrt{2} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

(ii) Same as (i) 😊

(7) Prove that a normal and nilpotent operator is zero operator.

Soln: Let T is normal and nilpotent
 Since T is nilpotent $\Rightarrow$ All e.v. of T are 0
 Since T normal $\Rightarrow T$ is diagonalizable
 Then $T = 0$. $\square$

Some question

Q A be an $n \times n$ diagonal matrix and
 $\chi_A(x) = (x - c_1)^{d_1} (x - c_2)^{d_2} \dots (x - c_k)^{d_k}$.

where $c_1, c_2, \dots, c_k$ are distinct. Let V be the space of $n \times n$ matrices B s.t.
 $AB = BA$. Prove $\dim V = d_1^2 + d_2^2 + \dots + d_k^2$

Soln: Let $B = \begin{bmatrix} B_{11} & \dots & B_{1k} \\ \vdots & & \vdots \\ B_{k1} & \dots & B_{kk} \end{bmatrix}$

Apply $AB = BA$

$$\begin{bmatrix} c_1 I & & \\ & c_2 I & \\ & & \ddots \\ & & & c_k I \end{bmatrix} \begin{bmatrix} B_{11} & \dots & B_{1k} \\ \vdots & & \vdots \\ B_{k1} & \dots & B_{kk} \end{bmatrix} = \begin{bmatrix} B_{11} & \dots & B_{1k} \\ \vdots & & \vdots \\ B_{k1} & \dots & B_{kk} \end{bmatrix} \begin{bmatrix} c_1 I & & \\ & c_2 I & \\ & & \ddots \\ & & & c_k I \end{bmatrix}$$

$\Rightarrow B = \begin{bmatrix} B_{11} & & \\ & \ddots & \\ & & B_{kk} \end{bmatrix}$ After simplifying
 $B_{ij} = 0, i \neq j$

B_{ii} can be arbitrary. Thus $\dim V = d_1^2 + \dots + d_k^2$

Q// Let $A = \begin{bmatrix} 0 & 0 & c \\ 1 & 0 & b \\ 0 & 1 & a \end{bmatrix}$

To show: $\chi_A(x) = \min_A(x) = x^3 - ax^2 - bx - c$

Soln: See, $\chi_A(x)$ 1 degree $\nexists$ $\exists$ $\eta \in \mathbb{R}$ s.t. $\min_A \neq 0$

Case ii) 2-degree pol, $\min_A(x) = a'x^2 + b'x + c'$

but $\min_A(A) = a'A^2 + b'A + c' \neq 0$

So it can be 2 degree.

Thus. $\min(A) = x^3 - ax^2 - bx - c$

Q// Find 3x3 matrix s.t. minimal pol. is x^3

Soln: $A = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Q// Let $V = P_n(\mathbb{R})$ (polynomials of at most degree n)

Let D be a diff. operator. then what is the minimal pol. of D .

Sol

$V = \text{span}\{1, x, x^2, \dots, x^n\} = \beta$ standard basis.

$D(1) = 0$

$D(x) = 1$

$D(x^2) = 2x$

$D(x^n) = n x^{n-1}$

$[D]_\beta = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 2 & \dots & 0 \\ 0 & 0 & 0 & 3 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 & n \end{bmatrix}$

(clear) $D^{n+1} = 0$, $D^n \neq 0$

So $\min_x(D) = x^{n+1}$

Q// $P(x,y) = (x,y)$ on $\mathbb{R}^2$
what is the min pol. of P .

Soln: $P(e_1) = e_1 = (1,0)$ $\beta = \{e_1, e_2\}$
 $P(e_2) = (0,0)$

$[P]_\beta = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$\chi_P(x) = x(x-1)$

$\min_P(x) = x(x-1)$

Q// Let A, B be $n \times n$ matrices over the field F . Do they have same characteristic polynomial? Do they have same minimal polynomial?

Ans: We have

$$\begin{aligned} \begin{vmatrix} I_n & A \\ B & \lambda I_n \end{vmatrix} &= \begin{vmatrix} I_n & A \\ B & \lambda I_n \end{vmatrix} \begin{vmatrix} I_n & -A \\ 0 & I_n \end{vmatrix} \\ &= \begin{vmatrix} I_n & 0 \\ B & \lambda I_n - BA \end{vmatrix} \\ &= |\lambda I_n - BA| \\ &= |BA - \lambda I_n| \end{aligned}$$

Similarly $\begin{vmatrix} I_n & A \\ B & \lambda I_n \end{vmatrix} = \begin{vmatrix} I_n & A \\ B & \lambda I_n \end{vmatrix} \begin{vmatrix} I_n & 0 \\ -B\lambda^{-1} & I_n \end{vmatrix}$

$$\begin{aligned} &= \begin{vmatrix} I_n - \lambda^{-1} AB & A \\ 0 & \lambda I_n \end{vmatrix} \\ &= |\lambda I_n - AB| \end{aligned}$$

□

Q// $V = M_{n \times n}(F)$

A be a fixed $n \times n$ matrix over F

$U(B) = AB - BA$
If A is diagonalizable then U is diagonalizable?

Soln: False, Let $F = \mathbb{R}$

$$\begin{aligned} A &= \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \\ B &= \begin{pmatrix} a & 1 \\ 1 & b \end{pmatrix} \end{aligned}$$

then $U(B) = AB - BA$

$$\begin{aligned} &= \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} a & 1 \\ 1 & b \end{pmatrix} - \begin{pmatrix} a & 1 \\ 1 & b \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \end{aligned}$$

$U(B)$ is not diagonalizable in $F = \mathbb{R}$

□

Q//